

Library Intelligence & Recommendation Database

SQL (MySQL/PostgreSQL)

Database Design

Recommendation System

Project Objective

An industry-oriented SQL project designed to manage library operations while leveraging data analytics to drive user engagement. This system automates inventory tracking, handles complex borrowing transactions, and implements a recommendation engine based on user reading patterns and content similarity.

Key Features



Automated Inventory Management: Real-time tracking of book availability using SQL Triggers. Automated stock decrement on borrow and increment on return.



Multi-Tier Recommendation Engine: Logic-driven suggestions including Content-Based (Favorite Genres) and Popularity-Based (Highest Rated) systems.



Late Fee & Overdue Logic: Advanced queries to identify delinquent accounts and calculate return durations for administrative oversight.

Database Schema

The database is built on a **Normalized Relational Model** consisting of 6 core entities:

- **Books & Authors:** Master data for library assets.
- **Categories:** Hierarchical genre classification.
- **Members:** Detailed user profile management.
- **Borrow_Records:** The transaction engine (Junction table).
- **Reviews:** Qualitative feedback loop for the recommendation logic.

Tech Stack

- ♦ **Language:** Structured Query Language (SQL)
- ♦ **Environment:** MySQL Workbench / PostgreSQL PgAdmin

Implementation & Key Queries

1. Recommendation Logic (Content-Based)

Identifying the favorite genre of a specific user and recommending unread books within that category:

```
WITH UserTopGenre AS (  
  SELECT category_id FROM Borrow_Records  
  JOIN Books ON Borrow_Records.book_id = Books.book_id  
  WHERE member_id = 101  
  GROUP BY category_id ORDER BY COUNT(*) DESC LIMIT 1  
)  
SELECT title FROM Books WHERE category_id = (SELECT * FROM UserTopGenre);
```

2. Analytics Dashboard Queries

- **Popularity:** Ranking books by borrow frequency.
- **Retention:** Identifying members with >5 borrows in 30 days.
- **Risk:** Tracking books not returned within the 14-day window.

GitHub Repository Structure

```
/scripts — Contains schema.sql and data_insertion.sql  
/docs — Detailed Database Diagram (ERD) and Documentation  
/analytics — SQL scripts for Business Intelligence insights  
README.md — Project overview and setup instructions
```

How to Run Locally

1. Clone the repository: `git clone [repo-url]`
2. Import `schema.sql` into your SQL environment.
3. Execute `data_insertion.sql` to populate sample records.
4. Run the queries in `insights.sql` to see the system in action.